

University of Information Technology & Sciences (UITS)**Faculty of Science and Engineering****Department of Computer Science and Engineering****Program: B.Sc. in CSE****Term Final Examination, Autumn 2024****Course Title: Coordinate Geometry and Vector Analysis****Course Code: MAT 261****Marks: 50****Time: 3(three) hours****(Answer all questions)**

Q.No.	Questions	Marks
1. a)	Define circle. Find the equation of the circle which passes through (3,5) and (5,-3) and has its centre on the line $2x + y = 27$.	[04]
b)	Reduce the equation to the standard form $32x^2 + 52xy - 7y^2 - 64x - 52y - 148 = 0$	[06]
2. a)	Find the direction cosines l, m, n of two lines which are connected by the relations, $l + m + n = 0, \quad l^2 + m^2 + n^2 = 0$	[05]
b)	Find the equation of the plane through the points (2,3,1), (1,1,3) and (2,2,3). Also find the distance from the point (5,0,-3) to this plane.	[05]
3. a)	The following forces act on a particle P. Find the unit vector parallel to the resultant vector of $\vec{r}_1 = 5\hat{i} + 3\hat{j} - 2\hat{k}$ and $\vec{r}_2 = -5\hat{i} + \hat{j} + 6\hat{k}$.	[03]
b)	Find the directional derivatives of ϕ at the point P(1,3,-2) in the direction of $2\hat{i} + \hat{j} - \hat{k}$.	[04]
c)	Find the projection of $A = 8\hat{i} - 2\hat{j} + \hat{k}$ on the vector $B = \hat{i} - \hat{j} + \hat{k}$.	[03]
4. a)	Define Gradient, Divergent and Curl. Suppose $A = x^2z^2\hat{i} - 2y^2z^2\hat{j} + xy^2z\hat{k}$. Find $\nabla \cdot (\nabla \times A)$ at the point P(1,1,1).	[06]
b)	A particle moves along the curve $x = 6t^3 + 2t, y = t^2 - 4t, z = 3t + 5$, where t is the time. Find the components of its velocity and acceleration at time $t = 3$.	[04]
5. a)	Suppose $A = (3x^2 + 6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k}$. Evaluate $\oint A \cdot d\mathbf{r}$ from (0,0,0) to (1,1,1) along the path where $x = t, y = t^2, z = t^3$.	[04]
b)	State Green's Theorem. Verify Green's theorem in the plane for $\oint (xy + y^2) dx + x^2 dy$ where C is the closed curve of the region bounded by $y = x$ and $y = x^2$.	[06]

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